

T.Y.B.Sc. COMPUTER SCIENCE

SEMESTER - VI

Lecture Notes

Subject Code: USCS601

Subject Name: Data Science

(NEW SYLLABUS W.E.F. 2023-2024)

Course Writer:

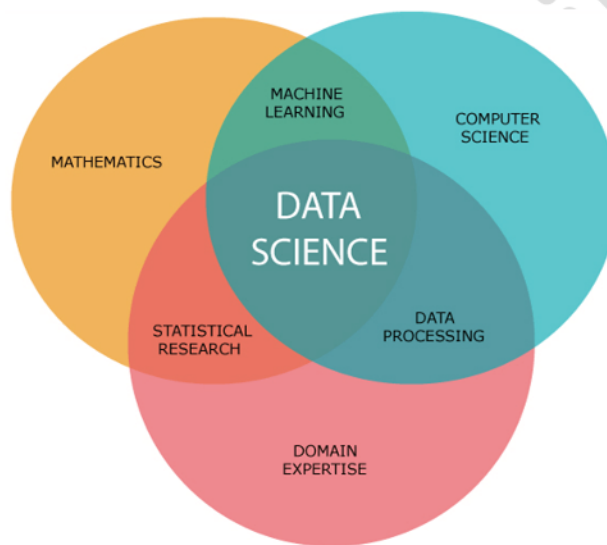
Prof. Hasan Phudinawala

Chapter No: 01

What is Data Science?

Definition

- Data science is a deep study of the massive amount of data, which involves extracting meaningful insights from raw, structured, and unstructured data that is processed using the scientific method, different technologies, and algorithms.
- Data science is an interdisciplinary field that uses algorithms, procedures, and processes to examine large amounts of data to uncover hidden patterns, generate insights, and direct decision-making.
- Data science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data.



Opportunity / Job roles

1. Data Scientist
2. Data Analyst
3. Machine learning expert
4. Data engineer
5. Data Architect
6. Data Administrator
7. Business Analyst
8. Business Intelligence Manager

Scope of Data Science

Every industry can collect and use data to identify patterns and make more informed business decisions. Below, you can learn how industries all over India use data science, including financial services, retail, health care,

agriculture, and utilities. However, this only represents a selection of the many uses of data science.

- **Financial services:** Data science is important in financial services for detecting potential fraud, evaluating how credit-worthy borrowers are, and predicting the performance of stocks or other financial products.
- **Retail:** In retail or e-commerce sectors, you can use data to personalise marketing or other experiences to your customers' needs. Data science can power recommendation engines and create targeted marketing strategies.
- **Health care:** Data science gives medical researchers and healthcare professionals the tools for faster research, more accurate treatment, and improved patient outcomes. Data science can even help read medical images more accurately using machine learning algorithms and computer vision.
- **Agriculture:** In agriculture, you can use data science to build models that use past events to predict weather patterns and crop yields. Farmers can also use data science to understand better what crops grow in what soil or what soil amendments they should make to create a more suitable environment for the crops they want to grow.
- **Utilities:** Energy and utility companies can use data science to understand how much energy their customers need or how energy consumption varies according to the weather and other patterns.

Applications & Domain of Data Science

1. In Search Engines

2. In Transport: For Example, In Driverless Cars the training data is fed into the algorithm and with the help of Data Science techniques, the Data is analyzed like what as the speed limit in highways, Busy Streets, Narrow Roads, etc. And how to handle different situations while driving etc.

3. In Finance: For Example, In Stock Market, Data Science is the main part. In the Stock Market, Data Science is used to examine past behavior with past data and their goal is to examine the future outcome. Data is analyzed in such a way that it makes it possible to predict future stock prices over a set timetable.

4. In E-Commerce: E-Commerce Websites like Amazon, Flipkart, etc. uses data Science to make a better user experience with personalized recommendations.

For Example, When we search for something on the E-commerce websites we get suggestions similar to choices according to our past data and also we get recommendations according to most buy the product, most rated, most searched, etc. This is all done with the help of Data Science.

5. In Health Care

In the Healthcare Industry data science act as a boon. Data Science is used for:

- Detecting Tumor.
- Drug discoveries.
- Medical Image Analysis.
- Virtual Medical Bots.
- Genetics and Genomics.
- Predictive Modeling for Diagnosis etc.

6. Image Recognition

When we upload our image with our friend on Facebook, Facebook gives suggestions Tagging who is in the picture. When an Image is Recognized, the data analysis is done on one's Facebook friends and after analysis, if the faces which are present in the picture matched with someone else profile then Facebook suggests us auto-tagging.

7. Targeting Recommendation

Suppose I want a mobile phone, so I just Google search it and after that, I changed my mind to buy offline. In Real -World Data Science helps those companies who are paying for Advertisements for their mobile. So everywhere on the internet in the social media, in the websites, in the apps everywhere I will see the recommendation of that mobile phone which I searched for. So this will force me to buy online.

8. Airline Routing Planning

With the help of Data Science, Airline Sector is also growing like with the help of it, it becomes easy to predict flight delays. It also helps to decide whether to directly land into the destination or take a halt in between like a flight can have a direct route from Delhi to the U.S.A or it can halt in between after that reach at the destination.

The airline industry uses data science to make flying safer and more efficient and to help the airline make more money by predicting delays, competitive discounts, and ticket pricing at different times of the year. Data science helps make flying safer because analysts can look at data from the past to set the best airplane routes for the future, which can also help determine when the best time to refuel airplanes is.

9. In Delivery Logistics

Various Logistics companies like DHL, FedEx, etc. make use of Data Science. Data Science helps these companies to find the best route for the Shipment of their Products, the best time suited for delivery, the best mode of transport to reach the destination, etc.

Q. Difference Between Data Science and Business Intelligence

S. No.	Factor	Data Science	Business Intelligence
1.	Concept	It is a field that uses mathematics, statistics and various other tools to discover the hidden patterns in the data.	It is basically a set of technologies, applications and processes that are used by the enterprises for business data analysis.
2.	Focus	It focuses on the future.	It focuses on the past and present.
3.	Data	It deals with both structured as well as unstructured data.	It mainly deals only with structured data.
4.	Flexibility	Data science is much more flexible as data sources can be added as per requirement.	It is less flexible as in case of business intelligence data sources need to be pre-planned.
5.	Method	It makes use of the scientific method.	It makes use of the analytic method.
6.	Complexity	It has a higher complexity in comparison to business intelligence.	It is much simpler when compared to data science.
7.	Expertise	It's expertise is data scientist.	It's expertise is the business user.
8.	Questions	It deals with the questions of what will happen and what if.	It deals with the question of what happened.
9.	Storage	The data to be used is disseminated in real-time clusters.	Data warehouse is utilized to hold data.
10.	Integration of data	The ELT (Extract-Load-Transform) process is generally used for the	The ETL (Extract-Transform-Load) process is generally used for the integration of data for

S. No.	Factor	Data Science	Business Intelligence
		integration of data for data science applications.	business intelligence applications.
11.	Tools	It's tools are SAS, BigML, MATLAB, Excel, etc.	It's tools are InsightSquared Sales Analytics, Klipfolio, ThoughtSpot, Cyfe, TIBCO Spotfire, etc.
12.	Usage	Companies can harness their potential by anticipating the future scenario using data science in order to reduce risk and increase income.	Business Intelligence helps in performing root cause analysis on a failure or to understand the current status.
13.	Business Value	Greater business value is achieved with data science in comparison to business intelligence as it anticipates future events.	Business Intelligence has lesser business value as the extraction process of business value carries out statically by plotting charts and KPIs (Key Performance Indicator).
14.	Handling data sets	The technologies such as Hadoop are available and others are evolving for handling understandingItsItsarge data sets.	The sufficient tools and technologies are not available for handling large data sets

Q. Difference Between Data Science and Artificial Intelligence

Aspects	Data science	Artificial Intelligence
Basics	Data Science is a detailed process that mainly involves pre- processing analysis, visualization and prediction.	AI(short) is the implementation of a predictive model to forecast future events and trends.
Goals	Identifying the patterns that are concealed in the data is the main objective of data science.	Automation of the process and the granting of autonomy to the data model are the main goals of artificial intelligence.
Types of data	Data Science will have a variety of different types of data, including structured, semi-structured, and unstructured type of data.	AI uses standardized data in the form of vectors and embeddings.
Scientific Processing	It has a high degree of scientific processing.	It has a lot of high levels of complex processing.
Tools used	The tools utilized in Data Science are far more extensive than those used in AI. This is due to the fact that Data Science entails a number of procedures for analyzing data and developing insights from it.	The tools used in AI are less extensive compared to Data Science.
Build	By using the concept of data science, we can build complex models about statistics and facts about data.	By using this we emulate cognition and human understanding to a certain level.
Technique used	It uses the technique of data analysis and data analytics.	It uses a lot of machine learning techniques.
Use	Data science makes use of graphical representation.	Artificial intelligence makes use of algorithms and network node representation.

Aspects	Data science	Artificial Intelligence
Knowledge	Its knowledge was established to find hidden patterns and trends in the data.	Its knowledge is all about imparting some autonomy to a data model.
Examples of Tools	R, Python, etc. are the tools used in data science.	Tensor flow, sci-kit-learn, Kaffee, etc are the tools used in AI.
Models	Models are built in Data Science to generate statistical insights for decision-making.	Models are created in Artificial Intelligence that is believed to be analogous to human understanding and cognition.
Applications	Its applications are advertising, marketing, Healthcare, etc.	Its application is robotics, automation, etc.

Q. Difference Between Data Science and Machine Learning

Data Science	Machine Learning
It deals with understanding and finding hidden patterns or useful insights from the data, which helps to take smarter business decisions.	It is a subfield of data science that enables the machine to learn from the past data and experiences automatically.
It is used for discovering insights from the data.	It is used for making predictions and classifying the result for new data points.
It is a broad term that includes various steps to create a model for a given problem and deploy the model.	It is used in the data modeling step of the data science as a complete process.
A data scientist needs to have skills to use big data tools like Hadoop, Hive and Pig, statistics, programming in Python, R, or Scala.	Machine Learning Engineer needs to have skills such as computer science fundamentals, programming skills in Python or R, statistics and probability concepts, etc.
It can work with raw, structured, and unstructured data.	It mostly requires structured data to work on.
Data scientists spent lots of time in handling the data, cleansing the data, and understanding its patterns.	ML engineers spend a lot of time for managing the complexities that occur during the implementation of algorithms and mathematical concepts behind that.

Q. Difference Between Data Science and Data Mining

Data Mining	Data Science
Data mining is a process of extracting useful information, patterns, and trends from huge databases.	Data science refers to the process of obtaining valuable insights from structured and unstructured data by using various tools and methods.
Data mining is a technique.	Data science is a field.
Primarily used for business purposes.	Primarily used for scientific purposes.
It is involved with the process.	It emphasizes the science of the data.
Data mining aims to make data more important and usable; it means extracting only useful information.	The objective of data science is to create a dominant data product.
Data mining is a technique that is a part of KDD (Knowledge discovery in database process).	It is related to the field of study like Mechanical engineering, Cloud architecture, etc.
It primarily deals with structured data.	It deals with any kind of data like structured, semi-structured, and unstructured.

Define Data Warehouse

- A **data warehouse** is a repository of information collected from multiple sources, stored under a unified schema, and usually residing at a single site.
- Data warehousing provides architectures and tools for business executives to systematically **organize, understand, and use** their data to make strategic decisions.
- A **data warehouse** is a **subject-oriented, integrated, time-varying, non-volatile** collection of data in support of the management's decision-making process.
- **Subject-oriented:** A data warehouse is organized around major subjects such as customer, products, sales, etc. Data are organized according to subject instead of application. For example, an insurance company using a data warehouse would organize their data by customer, premium, and claim instead of by different products (auto, life, etc.). The data organized by subject contains only the information necessary for the decision support processing.
- **Non-volatile:** A data warehouse is always a physically separate store of data, which is transformed from the application data found in the appropriate environment. Due to this separation, data warehouses do not require transaction processing, recovery, concurrency control, etc. The data are not updated or changed in any way once they enter the data warehouse, but are only loaded, refreshed and accessed for queries.
- **Time-varying:** Data are stored in a data warehouse to provide a historical perspective. Every key structure in the data warehouse contains, implicitly or explicitly, an element of time. The data warehouse contains a place for sorting data that are 5 to 10 years old, or older, to be used for comparisons, trends and forecasting.
- **Integrated:** A data warehouse is usually constructed by integrating multiple, heterogeneous sources such as relational databases, flat files, and OLTP files. When data resides in many separate applications in the operational environment, the encoding of data is often inconsistent. When data are moved from operational environment into the data warehouse, they assume a consistent coding convention.

What is Data Mining?

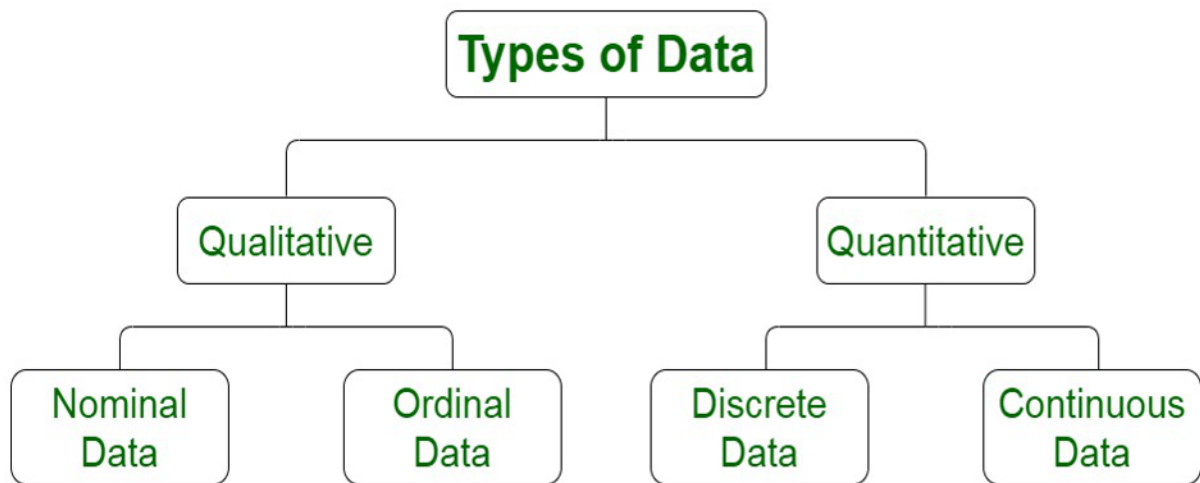
- **Data mining** is the process of discovering interesting patterns and knowledge from large amounts of data. The data sources can include databases, data warehouses, the Web, other information repositories, or data that are streamed into the system dynamically.
- **Data mining** defines extracting or mining knowledge from huge amounts of data. Data mining is generally used in places where a huge amount of data is saved and processed.
- **Data mining** is a collection of techniques for efficient automated discovery of previously unknown, valid, novel, useful and understandable patterns in large databases. The patterns must be actionable so they may be used in an enterprise's decision making.

Need for Data Mining

- Some facts about this rapid growth of data are as follows.
 - a) 48 hours of new videos are uploaded by YouTube users every minute of the day.
 - b) 571 new websites are created every minute of the day.
 - c) 90% of world's data has been produced in the last two years.
 - d) The production of data will be 44 times greater in 2020 than it was in 2010.
 - e) Brands and organizations on Facebook receive 34,722 'Likes' every minute of the day.
 - f) 100 terabytes of data is uploaded daily to Facebook.
 - g) In early 2012, there were more than 465 million accounts and 175 million tweets were shared by people on Twitter every day, according to Twitter's own research.
 - h) Every month, 30 billion pieces of content are shared by people on Facebook.
 - i) By late 2011, approximately 1.8 zetta byte of data had been created in that year alone, according the report published by IDC Digital Universe.

Chapter No: 02

Data Types and Sources



What is Structured Data?

Structured data is generally stored in tables in the form of rows and columns. Structured data in these tables can form relations with another tables. Humans and machines can easily retrieve information from structured data. This data is meaningful and is used to develop data models.

- Structured data is used by many business organizations. Companies apply data visualization techniques on the structured data to extract some meaningful insights from that data and develop data models.
- Data present in a Relational Database is the best example for structured data and this data can be accessed using a structured query language (SQL).
- Structured data is highly secured and requires low storage space. About 20% of the data is structured. Tools used on structured data are MySQL, PostgreSQL, SQLite, etc.

What is Unstructured Data?

Unprocessed and unorganized data is known as unstructured data. This type of data has no meaning and is not used to develop data models. Unstructured data may be text, images, audio, videos, reviews, satellite images, etc. Almost 80% of the data in this world is in the form of unstructured data.

- Unstructured data needs a lots of storage space. Here, data is not secured. It is difficult to search this data as it is not organized properly. This data is stored in NoSQL databases as they can't be managed using relational databases. It is very difficult to get insights from this data.

- Text files, Emails, data from social media applications, IoT, media etc., are examples of human generated unstructured data. Satellite images, scientific data etc., are examples of machine generated unstructured data.
- Tools used on unstructured data are MongoDB, Hadoop, DynamoDB, Azure, etc. Data visualization is best for analyzing unstructured data as they show hidden meaning of that data.

What is Semi-Structured Data?

Semi structured data is organized up to some extent only and the rest is unstructured. Hence, the level of organizing is less than that of Structured Data and higher than that of Unstructured Data.

- Semi-structured data is partially organized by means of XML/RDF.
- In semi-structured data, transaction management is not by default but is get adapted from DBMS, however there is no data concurrency.
- Data versioning is done only where tuples or graph is possible because semi structured data supports partial database.
- Semi-structured data is more flexible than structured data but less flexible and scalable as compared to unstructured data.
- If there is semi-structured data, then we can query only anonymous nodes, so its performance is lower than structured data but more than that of unstructured data.

Differences between Structured Data and Unstructured Data

Structured Data	Unstructured Data
Structured data is processed and organized.	Unstructured data is not processed and unorganized.
Data is stored in the form of tables.	Data is stored in the form of text, images etc.,
Structured data is managed using Relational database management system (RDBMS)	Unstructured data is managed using NoSQL
Data is highly secured.	Data is not secured.
Data models can be developed from structured data	We can't develop data models using unstructured data.
This data is stored in Data warehouses and Data lakes. It requires less storage space.	Unstructured data can be stored only in Data lakes. More storage is required to store this type of data.
Structured data is quantitative data.	Unstructured data is qualitative data.
Analytical methods used are: Classification, Regression, Clustering	Analytical methods used here are: Data stacking and Data mining
Searching is easy in this data	It is difficult to search as the data is not organized
Around 20% of the data is in structured form.	About 80% of the data is in unstructured form
As storage required is less, structured data is highly scalable	It is not scalable as it needs more storage
Data is not flexible.	Data is flexible
Example – Names, contact details, etc., are examples of structured data. Excel spreadsheets, Google sheets, relational databases contain structured data.	Example – Social media reviews, satellite images, polling results, etc., are examples of unstructured data. Unstructured data is stored in non-relational database management systems.

Q. Compare Qualitative & Quantitative Data

Qualitative Data	Quantitative Data
Qualitative data uses methods like interviews, participant observation, focus on a grouping to gain collective information.	Quantitative data uses methods as questionnaires, surveys, and structural observations to gain collective information.
Data format used in it is textual. Datasheets are contained of audio or video recordings and notes.	Data format used in it is numerical. Datasheets are obtained in the form of numerical values.
Qualitative data talks about the experience or quality and explains the questions like 'why' and 'how'.	Quantitative data talks about the quantity and explains the questions like 'how much', 'how many'.
The data is analyzed by grouping it into different categories.	The data is analyzed by statistical methods.
Qualitative data are subjective and can be further open for interpretation.	Quantitative data are fixed and universal.

Q. Compare Nominal Vs Ordinary Data

Characteristics	Nominal data	Ordinal Data
Nature of Categories	Distinct and Discrete	Discrete and Distinct
Order/Ranking	No inherent order	Has a clear order or ranking
Numerical Values	No meaningful numerical values	No meaningful numerical values
Analysis Techniques	Frequency counts, percentages, bar charts	Ranking, median, non-parametric tests, ordered bar charts, ordinal regression
Example	Colors, gender, types of animals	School grades, education level, seniority level
Interpretation	Used for classification and grouping based on category	Used for assessing ordered preferences, hierarchy, or rankings

Data sources

- Databases
- Files
- APIs
- Web scraping
- Sensors
- Social media

1. Databases

- A database is an organized collection of structured information, or data, typically stored electronically in a computer system.
- A database typically requires a comprehensive database software program known as a database management system (DBMS).

- A DBMS serves as an interface between the database and its end users or programs, allowing users to retrieve, update, and manage how the information is organized and optimized.
- There are different kinds of databases, ranging from the most prevalent approach, the relational database, object-oriented databases, to a distributed database, cloud database, and NoSQL databases.
- Databases are used to store and manage large amounts of structured and unstructured data, and they can be used to support a wide range of activities, including data storage, data analysis, and data management.
- Relational databases: These databases store data in the form of tables, with rows representing records and columns representing fields. Relationships between data can be established using keys.
- **Structured Query Language (SQL)** is a programming language designed for managing and manipulating data stored in relational database management systems (RDBMS).
- It is used to create, modify, and delete database objects such as tables, indices, and users; to manipulate data in the database by inserting, updating, and deleting records; and to query the database to retrieve specific data or generate reports.
- It is widely used in the development of web-based applications, and it is supported by most RDBMSs, including MySQL, Oracle, and Microsoft SQL Server.

2. Files

- Data stored in files, which can be in various formats such as text files, CSV, Excel, Spreadsheets, and more.
- A file data source is a computer file that contains connection information for a data source. The information in the file allows users to connect to the data source, and can be shared between users or applications. File data sources are often stored in text files and can include spreadsheets, PDFs, images, audio, video, and text documents.

3. API

- An application programming interface (API) is code that enables two software programs to communicate. An API facilitates the exchange of data, features and functionalities between software applications.
- An API facilitates programmers with an efficient way to develop their software programs. Thus, API helps two programs or applications to communicate with each other by providing them with the necessary tools and functions. It takes the request from the user and sends it

to the service provider and then again sends the result generated from the service provider to the desired user.

- A developer extensively uses APIs in his software to implement various features by using an API call without writing complex codes for the same. We can create an API for an operating system, database system, hardware system, JavaScript file, or similar object-oriented files. Also, an API is similar to a GUI (Graphical User Interface) with one major difference. Unlike GUIs, an application program interface helps software developers to access web tools while a GUI helps to make a program easier to understand for users.
- APIs are the building blocks for the today's websites in which heavy data is transferred from the client to server and vice versa.
- There are three basic forms of API –

1. WEB APIs

- A Web API also called Web Services is an extensively used API over the web and can be easily accessed using the HTTP protocols.
- The four main types of web APIs are:
 - Open API
 - Partner API
 - Internal API
 - Composite API

2. LOCAL APIs

- In this type of API, the programmers get the local middleware services. TAPI (Telephony Application Programming Interface), and .NET are common examples of Local APIs.

3. PROGRAM APIs

- It makes a remote program appear to be local by making use of RPCs (Remote Procedural Calls). SOAP is a well-known example of this type of API.

Few other types of APIs:

- **SOAP** (SIMPLE OBJECT ACCESS PROTOCOL): It defines messages in XML format used by web applications to communicate with each other.
- **REST** (Representational State Transfer): It makes use of HTTP to GET, POST, PUT, or DELETE data. It is basically used to take advantage of the existing data.
- **JSON-RPC**: It uses JSON for data transfer and is a lightweight remote procedural call defining a few data structure types.

- **XML-RPC:** It is based on XML and uses HTTP for data transfer. This API is widely used to exchange information between two or more networks.
- What are REST APIs?
- REST stands for Representational State Transfer, and follows the constraints of REST architecture allowing interaction with RESTful web services. It defines a set of functions (GET, PUT, POST, DELETE) that clients use to access server data. The functions used are:
 - GET (retrieve a record)
 - PUT (update a record)
 - POST (create a record)
 - DELETE (delete the record)

4. Web scraping

- Web scraping is the process of using bots to extract content and data from a website. Unlike screen scraping, which only copies pixels displayed onscreen, web scraping extracts underlying HTML code and, with it, data stored in a database. The scraper can then replicate entire website content elsewhere.
- Web scraping is an automatic method to obtain large amounts of data from websites. Most of this data is unstructured data in an HTML format which is then converted into structured data in a spreadsheet or a database so that it can be used in various applications.
- Web scraping requires two parts, namely the crawler and the scraper. The crawler is an artificial intelligence algorithm that browses the web to search for the particular data required by following the links across the internet. The scraper, on the other hand, is a specific tool created to extract data from the website.
- Web Scrapers can extract all the data on particular sites or the specific data that a user wants. Ideally, it's best if you specify the data, you want so that the web scraper only extracts that data quickly.

For example, you might want to scrape an Amazon page for the types of juicers available, but you might only want the data about the models of different juicers and not the customer reviews. So, when a web scraper needs to scrape a site, first the URLs are provided. Then it loads all the HTML code for those sites and a more advanced scraper might even extract all the CSS and Javascript elements as well. Then the scraper obtains the required data from this HTML code and outputs this data in the format specified by the user. Mostly, this is in the form of an Excel spreadsheet or a CSV file, but the data can also be saved in other formats, such as a JSON file.

5. Sensors

- A sensor is a device that detects and responds to some type of input from the physical environment. The input can be light, heat, motion, moisture, pressure, or any number of other environmental phenomena. Sensors collect data from the environment or devices, providing valuable information for various applications and IOT projects. In the context of data science sensor data is valuable for IOT applications, environmental monitoring, health care manufacturing and more.
- sensors can be a data source because they detect and respond to physical input, generating data that can be used in many ways:
- Sensor data is the information collected by a sensor device when it detects and responds to physical input.
- Sensor data can be used to provide information to users, guide processes, or as input to other systems. It can also be used to monitor environmental conditions, optimize asset utilization, and enable preventive maintenance.
- Sensor data is an important part of the internet of things (IoT) and edge computing environments. It's used in many fields, including manufacturing, healthcare, retail, oil and gas, automotive, energy, transportation, logistics, agriculture, and smart cities.

Chapter No: 03

Data Preprocessing

- **Data cleansing** or data cleaning is the process of detecting and correcting corrupt or inaccurate records from a record set, table, or database and refers to identifying incomplete, incorrect, inaccurate or irrelevant parts of the data and then replacing, modifying, or deleting the dirty or coarse data.
- Data cleaning is a crucial step in the machine learning (ML) pipeline, as it involves identifying and removing any missing, duplicate, or irrelevant data. The goal of data cleaning is to ensure that the data is accurate, consistent, and free of errors, as incorrect or inconsistent data can negatively impact the performance of the ML model.
- Data cleaning is essential because raw data is often noisy, incomplete, and inconsistent, which can negatively impact the accuracy and reliability of the insights derived from it. Data cleaning involves the systematic identification and correction of errors, inconsistencies, and inaccuracies within a dataset, encompassing tasks such as handling missing values, removing duplicates, and addressing outliers.

Handling Missing Values

- You can't ignore missing data because many algorithms will not accept missing values
- As a first option, you can drop observations that have missing values, but doing this will drop or lose information, so be mindful of this before you remove it.
- As a second option, you can input missing values based on other observations; again, there is an opportunity to lose integrity of the data because you may be operating from assumptions and not actual observations.
- As a third option, you might alter the way the data is used to effectively navigate null values.

1. Missing Values in Numerical Columns

The first approach is to replace the missing value with one of the following strategies:

- **Replace it with a constant value.** This can be a good approach when used in discussion with the domain expert for the data we are dealing with.
- **Replace it with the mean or median.** This is a decent approach when the data size is small—but it does add bias.
- **Replace it with values by using information from other columns.**

2. Predicting Missing Values Using an Algorithm

- Another way to predict missing values is to create a simple regression model. The column to predict here is the Salary, using other columns in the dataset.
- If there are missing values in the input columns, we must handle those conditions when creating the predictive model. A simple way to manage this is to choose only the features that do not have missing values or take the rows that do not have missing values in any of the cells.

3. Missing Values in Categorical Columns

- Dealing with missing data values in categorical columns is a lot easier than in numerical columns. Simply replace the missing value with a constant value or the most popular category.
- This is a good approach when the data size is small, though it does add bias. For example, we have a column for Education with two possible values: High School and College. If there are more people with a college degree in the dataset, we can replace the missing value with College Degree:

Outliers

- Often, there will be one-off observations where, they do not appear to fit within the data you are analysing.
- Outlier detection is the process of detecting outliers, or a data point that is far away from the average, and depending on what we are trying to accomplish.
- If you have a legitimate reason to remove an outlier, like improper data-entry, doing so will help the performance of the data you are working with.
- Detecting and appropriately dealing with outliers is essential in data science to ensure that statistical analysis and machine learning models are not unduly influenced, and the results are accurate and reliable.
- If an outlier proves to be irrelevant for analysis or is a mistake, consider removing it.

Duplicates

- Remove unwanted observations from your dataset, including duplicate observations or irrelevant observations.
- Duplicate observations will happen most often during data collection. When you combine data sets from multiple places, scrape data, or receive data from clients or multiple departments, there are opportunities to create duplicate data.
- De-duplication is one of the largest areas to be considered in this process.
- **Handling Duplicates:** The simplest and most straightforward way to handle duplicate data is to delete it. This can reduce the noise and

redundancy in our data, as well as improve the efficiency and accuracy of your models. However, we need to be careful and make sure that you are not losing any valuable or relevant information by removing duplicate data.

- We also need to consider the criteria and logic for choosing which duplicates to keep or discard. For example, we can use the `df.drop_duplicates()` method in pandas to remove duplicate rows or columns, specifying the subset, keep, and inplace arguments. Removing duplicates: In python using Pandas: `df.drop_duplicates()`.

Data Transformation

- Data transformation is the process of converting data from one format or structure into another format or structure.
- It is a fundamental aspect of most data integration and data management tasks such as data wrangling, data warehousing, data integration and application integration.
- Data transformation is part of an ETL process and refers to preparing data for analysis and modelling. This involves cleaning (removing duplicates, fill-in missing values), reshaping (converting currencies, pivot tables), and computing new dimensions and metrics.
- Data transformation techniques include **scaling, normalization, and encoding categorical variables**.
- These techniques can help to improve model performance, reduce the impact of outliers, and ensure that the data is on the same scale.

1. Scaling

- Feature scaling is a data preprocessing technique used to transform the values of features or variables in a dataset to a similar scale.
- The purpose is to ensure that all features contribute equally to the model and to avoid the domination of features with larger values.
- Feature scaling becomes necessary when dealing with datasets containing features that have different ranges, units of measurement, or orders of magnitude.
- In such cases, the variation in feature values can lead to biased model performance or difficulties during the learning process.
- Scaling is the process of transforming the features of a dataset so that they fall within a specific range. Scaling is useful when we want to compare two different variables on equal grounds. This is especially useful with variables which use distance measures.
- There are several common techniques for feature scaling, including standardization, normalization, and min-max scaling. These methods adjust the feature values while preserving their relative relationships and distributions.

- **Algorithms that require scaled data:**

1. Gradient Descent Based Algorithms
2. Distance-Based Algorithms
3. Tree-Based Algorithms

2. Normalization

- Data normalization is a technique used in data mining to transform the values of a dataset into a common scale. This is important because many machine learning algorithms are sensitive to the scale of the input features and can produce better results when the data is normalized.
- There are several different normalization techniques that can be used in data mining, including:
- **Min-Max normalization:** This technique scales the values of a feature to a range between 0 and 1. This is done by subtracting the minimum value of the feature from each value, and then dividing it by the range of the feature.
- **Z-score normalization:** This technique scales the values of a feature to have a mean of 0 and a standard deviation of 1. This is done by subtracting the mean of the feature from each value, and then dividing it by the standard deviation.
- **Decimal Scaling:** This technique scales the values of a feature by dividing the values of a feature by a power of 10.
- **Logarithmic transformation:** This technique applies a logarithmic transformation to the values of a feature. This can be useful for data with a wide range of values, as it can help to reduce the impact of outliers.

3. Encoding Categorical Variables

- **Data Encoding** is an important pre-processing step in Machine Learning. It refers to the process of converting categorical or textual data into numerical format, so that it can be used as input for algorithms to process. The reason for encoding is that most machine learning algorithms work with numbers and not with text or categorical. Most machine learning algorithms work only with numerical data, so categorical variables (such as text labels) must be transformed into numerical values.

Types of Categorical Data

Categorical data can be broadly classified into two types:

1. **Nominal Data:** This type of data represents categories without any inherent order. Examples include gender (male, female), color (red, blue, green), and country (USA, India, UK).
2. **Ordinal Data:** This type of data represents categories with a meaningful order or ranking. Examples include education level (high

school, bachelor's, master's, PhD) and customer satisfaction (low, medium, high).

Encoding Techniques

Scikit-learn provides several methods to encode categorical data. Let's explore the most commonly used techniques:

1. Label Encoding

Label Encoding is a simple and straightforward method that assigns a unique integer to each category. This method is suitable for ordinal data where the order of categories is meaningful. **Use Case:** Applicable for cases in which the ordering of categories has analytical relevance.

Syntax:

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
data['encoded_column'] = le.fit_transform(data['categorical_column'])
```

Sample code for practice

```
from sklearn.preprocessing import LabelEncoder
data = ['red', 'blue', 'green', 'blue', 'red']
label_encoder = LabelEncoder()
encoded_data = label_encoder.fit_transform(data)
print(encoded_data)
```

One-Hot Encoding

- One-Hot Encoding converts categorical data into a binary matrix, where each category is represented by a binary vector. This method is suitable for nominal data. Use Case: Most appropriate for those situations, where the categories do not have an inherent order, or there is a clear distinction between them.
- In one hot categorical encoding, for each level of a categorical feature, we create a new variable. Each category is mapped with a binary variable containing either 0 or 1. Here, 0 represents the absence, and 1 represents the presence of that category.
- These newly created binary features are known as Dummy variables. The number of dummy variables depends on the levels present in the categorical variable. This might sound complicated. hot encode this data.

Syntax:

```
from sklearn.preprocessing import OneHotEncoder  
ohe = OneHotEncoder(sparse=False)  
encoded_data = ohe.fit_transform(data[['categorical_column']])
```

Sample code for practice

```
from sklearn.preprocessing import OneHotEncoder  
import numpy as np  
data = np.array(['red', 'blue', 'green', 'blue', 'red']).reshape(-1, 1)  
onehot_encoder = OneHotEncoder(sparse_output=False)  
encoded_data = onehot_encoder.fit_transform(data)  
print(encoded_data)
```

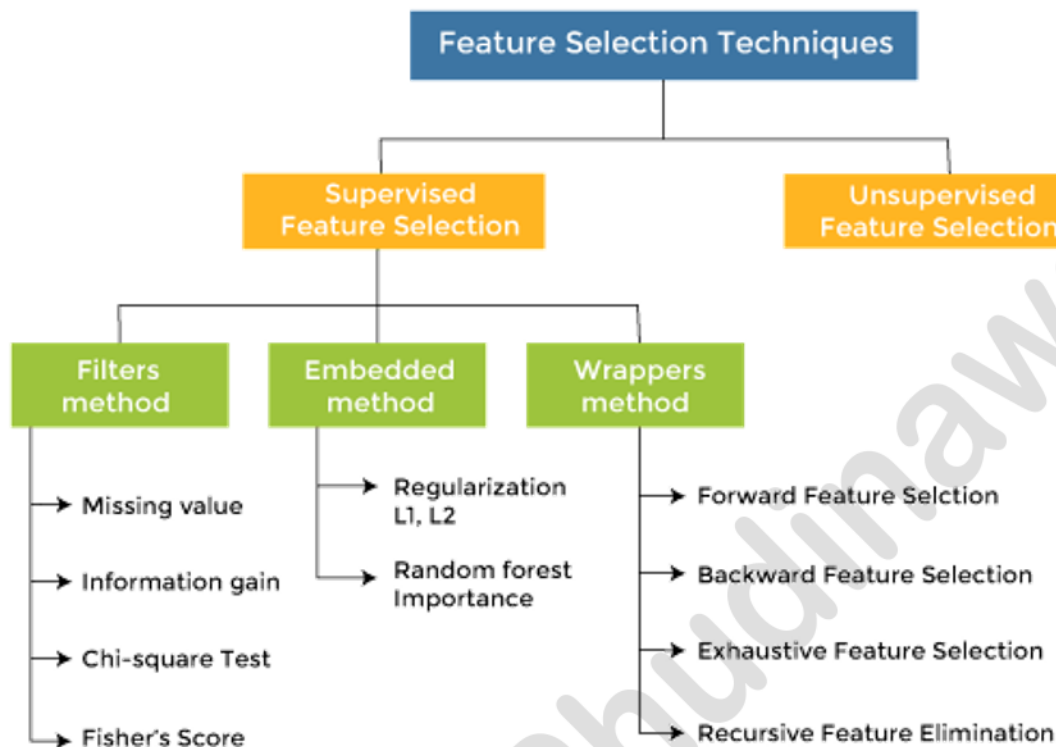
Feature selection

- ***It is a process of automatically or manually selecting the subset of most appropriate and relevant features to be used in model building.*** Feature selection is performed by either including the important features or excluding the irrelevant features in the dataset without changing them.
- feature selection and extraction processes may have the same objective, both are completely different from each other.
- The main difference between them is that feature selection is about selecting the subset of the original feature set, whereas feature extraction creates new features. Feature selection is a way of reducing the input variable for the model by using only relevant data in order to reduce overfitting in the model.

Below are some benefits of using feature selection in machine learning:

- It helps in avoiding the curse of dimensionality.
- It helps in the simplification of the model so that it can be easily interpreted by the researchers.
- It reduces the training time.
- It reduces overfitting hence enhance the generalization.

Feature Selection Techniques



1. Wrapper Methods

- In wrapper methodology, selection of features is done by considering it as a search problem, in which different combinations are made, evaluated, and compared with other combinations. It trains the algorithm by using the subset of features iteratively.
- On the basis of the output of the model, features are added or subtracted, and with this feature set, the model has trained again.

Some techniques of wrapper methods are:

- **Forward selection** - Forward selection is an iterative process, which begins with an empty set of features. After each iteration, it keeps adding on a feature and evaluates the performance to check whether it is improving the performance or not. The process continues until the addition of a new variable/feature does not improve the performance of the model.
- **Backward elimination** - Backward elimination is also an iterative approach, but it is the opposite of forward selection. This technique begins the process by considering all the features and removes the least significant feature. This elimination process continues until removing the features does not improve the performance of the model.

- **Exhaustive Feature Selection-** Exhaustive feature selection is one of the best feature selection methods, which evaluates each feature set as brute-force. It means this method tries & make each possible combination of features and return the best performing feature set.
- **Recursive Feature Elimination** – Recursive feature elimination is a recursive greedy optimization approach, where features are selected by recursively taking a smaller and smaller subset of features. Now, an estimator is trained with each set of features, and the importance of each feature is determined using *coef_attribute* or through a *feature_importances_attribute*.

2. Filter Methods

- In Filter Method, features are selected on the basis of statistics measures. This method does not depend on the learning algorithm and chooses the features as a pre-processing step.
- The filter method filters out the irrelevant feature and redundant columns from the model by using different metrics through ranking.
- The advantage of using filter methods is that it needs low computational time and does not overfit the data.

Some common techniques of Filter methods are as follows:

- **Information Gain:** Information gain determines the reduction in entropy while transforming the dataset. It can be used as a feature selection technique by calculating the information gain of each variable with respect to the target variable.
- **Chi-square Test:** Chi-square test is a technique to determine the relationship between the categorical variables. The chi-square value is calculated between each feature and the target variable, and the desired number of features with the best chi-square value is selected.
- **Fisher's Score:** Fisher's score is one of the popular supervised technique of features selection. It returns the rank of the variable on the fisher's criteria in descending order. Then we can select the variables with a large fisher's score.
- **Missing Value Ratio:** The value of the missing value ratio can be used for evaluating the feature set against the threshold value. The formula for obtaining the missing value ratio is the number of missing values in each column divided by the total number of observations. The variable is having more than the threshold value can be dropped.

3. Embedded Methods

- Embedded methods combined the advantages of both filter and wrapper methods by considering the interaction of features along with low computational cost. These are fast processing methods similar to the filter method but more accurate than the filter method.
- These methods are also iterative, which evaluates each iteration, and optimally finds the most important features that contribute the most to training in a particular iteration. Some techniques of embedded methods are:
- **Regularization**- Regularization adds a penalty term to different parameters of the machine learning model for avoiding overfitting in the model. This penalty term is added to the coefficients; hence it shrinks some coefficients to zero. Those features with zero coefficients can be removed from the dataset. The types of regularization techniques are L1 Regularization (Lasso Regularization) or Elastic Nets (L1 and L2 regularization).
- **Random Forest Importance** - Different tree-based methods of feature selection help us with feature importance to provide a way of selecting features. Here, feature importance specifies which feature has more importance in model building or has a great impact on the target variable. Random Forest is such a tree-based method, which is a type of bagging algorithm that aggregates a different number of decision trees. It automatically ranks the nodes by their performance or decrease in the impurity (Gini impurity) over all the trees. Nodes are arranged as per the impurity values, and thus it allows to pruning of trees below a specific node. The remaining nodes create a subset of the most important features.

Data merging: combining multiple datasets

- Data merging is the process of combining two or more similar records into a single one. Merging is done to add variables to a dataset, append or add cases or observations to a dataset, or remove duplicates and other incorrect information.
- Merging data into a single point is necessary in certain situations, especially when an organization needs to add new cases, variables, or data based on the lookup values. However, data merging needs to be performed with caution; otherwise, it can lead to duplication, inaccuracy, or inconsistency issues.

Data from multiple sources is merged in a number of scenarios:

- Digital transformation initiatives
- Driving business intelligence

- Integration after mergers and acquisitions, when data from different organizations are merged into one dataset
- Different applications including customer relationship management, marketing automation tools, and website analytics tools are merged for analysis, processing, and predictions.

Stages of data merging

The process of merging can be categorized into three stages: pre merging, merging, and post-merging processes.

1. Pre-merging process

- **Data Profiling:** Before merging, it is crucial to profile the data, analyzing the different parts of data sources. This step helps an organization understand the outcomes of merging and prevent any potential errors that may occur. Data profiling consists of two important steps:
- Analyzing the list of attributes that each data source possesses. This step helps an organization understand how the merged data will scale, what attributes are intended to be merged, and what may need to be supplemented.
- Analysis of the data values in each part of a source to assess the completeness, uniqueness, and distribution of attributes. In short, a data profile validates the attributes of a predefined pattern and helps to identify invalid values.

Standardize and Transform Data

- Data sources may contain incomplete and invalid values. These datasets cannot be merged before they are standardized. In addition to errors, data attributes from different sources may contain the same information, such as customer names. However, the format of these data values can be entirely different. Due to the lexical and structural differences in datasets, data loss and errors may occur. In order to standardize data, certain factors need to be managed.
- Invalid characters that cannot be printed, void values, and trailing spaces should be replaced with valid values. An example is not allowing more than one space when data is entered or reducing all multiple spaces to one when transforming data.
- To standardize long fields of data, records should be parsed into smaller parts in different source files. This helps to ensure data accuracy remains even after data sources are merged.
- Constraints for integration should be defined. For example, the maximum or minimum number of characters in a certain field should be defined, or a hyphenated surname should contain no spaces.

Data Filtering:

- A part or subset of the original data sources can be merged instead of an entire data source. Such horizontal slicing of the data is done when data in a constrained period of time needs to be merged, or when only a subgroup of rows meets the conditional criteria. Vertical slicing can only be done when a data source contains attributes that do not have any valuable information.

Data Uniqueness:

- Many times, the information from a single entity may be stored across a number of sources. Merging data becomes more complex if the datasets contain duplicates. Therefore, before beginning any merging process, it is important to run data matching algorithms. These help to apply conditional rules to identify and delete duplicates and result in the uniqueness of records across all sources.

2. Merging: Integration and aggregation of data

- The process of merging can either be an integration or an aggregation. Once all the previous steps have been completed, the data is ready for merging. There are a number of ways this process can be achieved.

Append Rows: Appending rows is done when records sourced from different datasets need to be combined in one place. The data sources to be joined need to have an identical structure. The data types, pattern validations, and integrity constraints of corresponding columns also need to be the same to avoid invalid formatting problems. Data matching should be performed with or before merging if the data of an entity is from different sources.

Appending Columns: This process is done when more dimensions need to be added to an existing record. In such scenarios, all columns from different sources must be made unique. Every single record should be uniquely identifiable across all sets of data, making it easier for records with the same identifier to be merged. If the merging column does not contain any data, then null values should be specified for all records from that dataset. However, if many datasets contain the same or highly similar dimension information, then these dimensions can be merged together in one field.

Conditional Merging: Conditional merging is used when there are incomplete datasets that need to be united. In this type of merge, values from one dataset need to be looked up and the other datasets need to be filled in accordingly. The source dataset from which values are looked up should contain all unique records. However, the dataset to which data is being appended or the target dataset does not need to have unique values.

3. Post-merging process

Once the merge process is finished, it is important to do a final profile audit of the merged source, as is done at the start of the merge process. This will help find errors that may have occurred during the merge. Any inaccurate, incomplete, or invalid values can also be spotted.

Prof. Hasan Phudinawala

Chapter No: 04

Data Wrangling and Feature Engineering

- A **data wrangling** process, also known as a data munging process, consists of reorganizing, transforming, and mapping data from one "raw" form into another to make it more usable and valuable for a variety of downstream uses including analytics.
- Data wrangling, or data munging, is a crucial process in the data analytics workflow that involves cleaning, structuring, and enriching raw data to transform it into a more suitable format for analysis.
- Also known as data cleaning or data munging, data wrangling enables businesses to tackle more complex data in less time, produce more accurate results, and make better decisions.

Reshaping:

- Reshaping data involves changing the structure of the dataset. The shape of a data set refers to the way in which a data set is arranged into rows and columns, and reshaping data is the rearrangement of the data without altering the content of the data set.
- Reshaping data sets is a very frequent and cumbersome task in the process of data manipulation and analysis.
- It is commonly used to move between "wide" and "long" data formats:
- **Wide format:** Each variable has its own column.
- **Long format:** Each observation of a variable is stored in a single row.

Methods

1. **.melt():** Converts wide-format data to long-format.
2. **.stack() / .unstack():** Converts between different levels of a MultiIndex.

Code 1:

```
import pandas as pd
# Wide format dataset
data = {
    'Year': [2020, 2021],
    'Q1_Sales': [150, 250],
    'Q2_Sales': [200, 300]
}
df = pd.DataFrame(data)
print("Wide Format Data:")
print(df)
# Convert to long format
long_df = df.melt(id_vars='Year', var_name='Quarter', value_name='Sales')
print("\nLong Format Data:")
print(long_df)
```

output:

```
Wide Format Data:
   Year  Q1_Sales  Q2_Sales
0  2020         150         200
1  2021         250         300

Long Format Data:
   Year  Quarter  Sales
0  2020   Q1_Sales    150
1  2021   Q1_Sales    250
2  2020   Q2_Sales    200
3  2021   Q2_Sales    300
```

code 2:

Convert columns into a MultiIndex (long format)

```
stacked = df.set_index('Year').stack()
```

```
print("\nStacked Data:")
```

```
print(stacked)
```

Convert back to wide format

```
unstacked = stacked.unstack()
```

```
print("\nUnstacked Data:")
```

```
print(unstacked)
```

Output:

```
Stacked Data:
Year
2020  Q1_Sales    150
      Q2_Sales    200
2021  Q1_Sales    250
      Q2_Sales    300
dtype: int64

Unstacked Data:
      Q1_Sales  Q2_Sales
Year
2020         150         200
2021         250         300
```

Pivoting:

Pivoting transforms data by reorganizing rows and columns based on values in specific columns. It's often used to summarize or display the data in a more readable format.

Methods

1. **.pivot():** Rearranges the data based on specific columns.
2. **.pivot_table():** Similar to .pivot(), but allows aggregation functions for duplicate entries

Code:

```
# Long format data
data = {
    'Year': [2020, 2020, 2021, 2021],
    'Quarter': ['Q1', 'Q2', 'Q1', 'Q2'],
    'Sales': [150, 200, 250, 300]
}
df = pd.DataFrame(data)
print("Original Long Format Data:")
print(df)
```

```
# Pivot to wide format
```

```
pivoted = df.pivot(index='Year', columns='Quarter', values='Sales')
print("\nPivoted Wide Format Data:")
print(pivoted)
```

Output:

Original Long Format Data:

	Year	Quarter	Sales
0	2020	Q1	150
1	2020	Q2	200
2	2021	Q1	250
3	2021	Q2	300

Pivoted Wide Format Data:

	Quarter	Q1	Q2
Year			
2020		150	200
2021		250	300

Aggregating:

Aggregation is the process of applying statistical or mathematical operations (like sum, mean, count) to summarize data. It's often performed after grouping data into categories.

Method

1. **.groupby():** Groups data by one or more columns.
2. **.agg():** Allows applying multiple aggregation functions.

Code 1:

```
data = {
    'Year': [2020, 2020, 2021, 2021],
    'Quarter': ['Q1', 'Q2', 'Q1', 'Q2'],
    'Sales': [150, 200, 250, 300]
}
df = pd.DataFrame(data)
print("DataFrame is :")
print(df)
# Group by year and sum sales
grouped = df.groupby('Year')['Sales'].sum()

print("\nGrouped Data (Sum of Sales by Year):")
print(grouped)
```

Output:

DataFrame is :

	Year	Quarter	Sales
0	2020	Q1	150
1	2020	Q2	200
2	2021	Q1	250
3	2021	Q2	300

Grouped Data (Sum of Sales by Year):

Year	
2020	350
2021	550

Name: Sales, dtype: int64

Code 1:

```
# Group by year and apply multiple aggregations
```

```
agg_data = df.groupby('Year').agg(  
    total_sales=('Sales', 'sum'),  
    avg_sales=('Sales', 'mean'),  
    count_quarters=('Quarter', 'count')  
)  
print("\nAggregated Data with Multiple Functions:")  
print(agg_data)
```

Output:

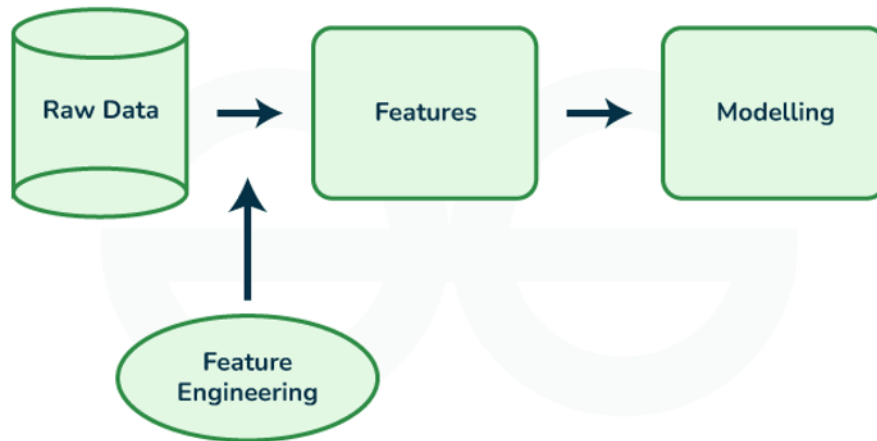
Aggregated Data with Multiple Functions:			
	total_sales	avg_sales	count_quarters
Year			
2020	350	175.0	2
2021	550	275.0	2

Feature engineering**What is a Feature?**

- In the context of machine learning, a feature (also known as a variable or attribute) is an individual measurable property or characteristic of a data point that is used as input for a machine learning algorithm.
- Features can be numerical, categorical, or text-based, and they represent different aspects of the data that are relevant to the problem at hand.
- For example, in a dataset of housing prices, features could include the number of bedrooms, the square footage, the location, and the age of the property. In a dataset of customer demographics, features could include age, gender, income level, and occupation.
- **Feature engineering** is the process of transforming raw data into features that are suitable for machine learning models.
- In other words, it is the process of selecting, extracting, and transforming the most relevant features from the available data to build more accurate and efficient machine learning models.
- The success of machine learning models heavily depends on the quality of the features used to train them.

Processes Involved in Feature Engineering

- Feature engineering in Machine learning consists of mainly 5 processes: Feature Creation, Feature Transformation, Feature Extraction, Feature Selection, and Feature Scaling.



Feature Creation

- Feature Creation is the process of generating new features based on domain knowledge or by observing patterns in the data. It is a form of feature engineering that can significantly improve the performance of a machine-learning model.

Types of Feature Creation:

- Domain-Specific:** Creating new features based on domain knowledge, such as creating features based on business rules or industry standards.
- Data-Driven:** Creating new features by observing patterns in the data, such as calculating aggregations or creating interaction features.
- Synthetic:** Generating new features by combining existing features or synthesizing new data points.

Why Feature Creation?

- Improves Model Performance:** By providing additional and more relevant information to the model, feature creation can increase the accuracy and precision of the model.
- Increases Model Robustness:** By adding additional features, the model can become more robust to outliers and other anomalies.
- Improves Model Interpretability:** By creating new features, it can be easier to understand the model's predictions.
- Increases Model Flexibility:** By adding new features, the model can be made more flexible to handle different types of data.

What is Feature Scaling?

- Feature Scaling is a technique to standardize the independent features present in the data in a fixed range. It is performed during the data pre-processing to handle highly varying magnitudes or values or units.
- If feature scaling is not done, then a machine learning algorithm tends to weigh greater values, higher and consider smaller values as the lower values, regardless of the unit of the values.
- There are several common techniques for feature scaling, including **standardization**, **normalization**, and **min-max scaling**.
- These methods adjust the feature values while preserving their relative relationships and distributions. By applying feature scaling, the dataset's features can be transformed to a more consistent scale, making it easier to build accurate and effective machine learning models.
- Scaling facilitates meaningful comparisons between features, improves model convergence, and prevents certain features from overshadowing others based solely on their magnitude.

Normalization: Normalization, a vital aspect of Feature Scaling, is a data preprocessing technique employed to standardize the values of features in a dataset, bringing them to a common scale. This process enhances data analysis and modeling accuracy by mitigating the influence of varying scales on machine learning models. Normalization is a scaling technique in which values are shifted and rescaled so that they end up ranging between 0 and 1. It is also known as Min-Max scaling.

$$X_{\text{new}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$$

Here, X_{max} and X_{min} are the maximum and the minimum values of the feature, respectively.

- When the value of X is the minimum value in the column, the numerator will be 0, and hence X' is 0.
- On the other hand, when the value of X is the maximum value in the column, the numerator is equal to the denominator, and thus the value of X' is 1
- If the value of X is between the minimum and the maximum value, then the value of X' is between 0 and 1.

Standardization: Standardization is another Feature scaling method where the values are centered around the mean with a unit standard deviation.

This means that the mean of the attribute becomes zero, and the resultant distribution has a unit standard deviation.

Here's the formula for standardization:

$$Z = \frac{x - \mu}{\sigma}$$

:

S.NO.	Normalization	Standardization
1.	Minimum and maximum value of features are used for scaling	Mean and standard deviation is used for scaling.
2.	It is used when features are of different scales.	It is used when we want to ensure zero mean and unit standard deviation.
3.	Scales values between [0, 1] or [-1, 1].	It is not bounded to a certain range.
4.	It is really affected by outliers.	It is much less affected by outliers.
5.	Scikit-Learn provides a transformer called MinMaxScaler for Normalization.	Scikit-Learn provides a transformer called StandardScaler for standardization.
6.	This transformation squishes the n-dimensional data into an n-dimensional unit hypercube.	It translates the data to the mean vector of original data to the origin and squishes or expands.
7.	It is useful when we don't know about the distribution	It is useful when the feature distribution is Normal or Gaussian.
8.	It is a often called as Scaling Normalization.	It is a often called as Z-Score Normalization.

Chapter No: 05

Tools and Libraries

Pandas

Pandas is an open-source Python library used for data manipulation and analysis. It provides data structures and functions needed to work efficiently with structured data, such as tables or time series. It is built on top of the NumPy library and is widely used in data science, machine learning, and scientific computing.

Key Features of Pandas:

- **Data Structures:** Provides two primary data structures:
 - **Series:** A one-dimensional array-like object.
 - **DataFrame:** A two-dimensional, tabular data structure with labeled axes (rows and columns).
- Tools to filter, group, merge, and reshape data.
- Easy loading and saving of data in various formats, such as CSV, Excel, SQL databases, JSON, etc.
- Provides functions to detect, remove, or fill missing values.
- Automatically aligns data in calculations.

Advantages of Pandas:

- **User-Friendly:** Simplifies data handling and makes it easy to perform complex data manipulations with intuitive syntax.
- **Versatile:** Handles a wide variety of data formats (structured, unstructured, time-series).
- **Flexible Indexing:** Supports labeled indexing, making data access and manipulation easier.
- **Rich Ecosystem:** Integrates seamlessly with other Python libraries like Matplotlib, NumPy, Scikit-learn, and Seaborn.
- **Visualization Support:** Works with plotting libraries to create graphs and visualizations directly from DataFrames.

Uses of Pandas:

- **Data Cleaning & Data Wrangling:**
- **Exploratory Data Analysis (EDA):** Quickly summarizing and exploring datasets using statistical and visualization techniques.

- **Time Series Analysis:** Handling time-indexed data and performing operations like resampling.
- **Financial Data Analysis:** Processing and analyzing large datasets from financial markets.
- **Machine Learning:** Preprocessing data, such as normalization, encoding categorical data, and feature engineering.

NumPy

- It is a fundamental library for numerical computing in Python, widely used in data analysis, scientific computing, and machine learning.
- It provides support for large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to perform efficient operations on these arrays.
- At its core is the ndarray object, a fast and flexible container for homogeneous data. NumPy leverages optimized C and Fortran backends for high-performance computations.
- It is a foundational library upon which many others, such as SciPy and Pandas, are built.
- Applications of NumPy include data preprocessing, linear algebra, random number generation, and statistical operations, making it indispensable for numerical tasks in Python.

Scikit-learn

- It is a widely used machine learning library built on top of NumPy, SciPy, and Matplotlib. It provides simple and efficient tools for a variety of machine learning tasks, including supervised and unsupervised learning.
- Scikit-learn offers **algorithms** for classification, regression, clustering, and dimensionality reduction, as well as utilities for model selection, cross-validation, and performance evaluation.
- The library also includes **preprocessing tools** for scaling, normalizing, and encoding data. Its core components include estimators for building and fitting models, transformers for data preprocessing, and pipelines for integrating multiple steps into streamlined workflows.
- Scikit-learn is extensively used in predictive modeling, feature engineering, and data analysis across industries such as healthcare, finance, and marketing.

Unit No: II

Chapter No: 06

Exploratory Data Analysis (EDA)

- Exploratory Data Analysis (EDA) refers to the method of studying and exploring record sets to apprehend their predominant traits, discover patterns, locate outliers, and identify relationships between variables.
- EDA is normally carried out as a preliminary step before undertaking extra formal statistical analyses or modeling.
- EDA, or Exploratory Data Analysis, refers back to the method of analyzing and analyzing information units to uncover styles, pick out relationships, and gain insights.
- EDA involves both graphical displays of data and numerical summaries of data.

Descriptive Statistics

- Descriptive statistics refers to a set of methods used to summarize and describe the main features of a dataset, such as its central tendency, variability, and distribution.
- These methods provide an overview of the data and help identify patterns and relationships
- Descriptive statistics summarize and describe the main features of a dataset.

a. Mean

- The arithmetic average of a dataset.
- Arithmetic mean is the sum of data divided by the number of data-points. It is a measure of the central location of data in a set of values which vary in range. In Python, we usually do this by dividing the sum of given numbers with the count of number present.

Given set of numbers : [n1, n2, n3, n5, n6]

Sum of data-set = (n1 + n2 + n3 + n4 + n5)

Number of data produced = 5

Average or arithmetic mean = (n1 + n2 + n3 + n4 + n5) / 5

- **Python Example:**

```
import numpy as np
data = [10, 20, 30, 40, 50]
mean = np.mean(data)
print("Mean:", mean)
```


b. Median

- The median is another measure of central tendency in a dataset. • The median is the middle value of a dataset when it is ordered from least to greatest.
- If the dataset has an even number of values, the median is the average of the two middle values.
- Here's how to find the median:
 - Sort the Data: Arrange the values in the dataset in ascending order.
 - Identify the Middle Value:
 - If the dataset has an odd number of values, the median is the middle value.
- If the dataset has an even number of values, the median is the average of the two middle values. The middle value when data is sorted.
- For even-sized datasets, it's the average of the two middle numbers.

- **Python Example:**

```
median = np.median(data)
print("Median:", median)
```

c. Mode

- The mode is another measure of central tendency used in statistics. • The mode represents the most frequently occurring value(s) in the dataset.
- A dataset may have no mode, one mode (unimodal), or more than one mode (multimodal).
- Here's how to find the mode:
 1. Identify Values and Frequencies: Determine the frequency of each distinct value in the dataset.
 2. Identify Modes: The mode(s) are the value(s) with the highest frequency. If no value is repeated, the dataset is considered to have no mode.
- The most frequently occurring value(s) in a dataset.
- For example, consider the dataset {3, 5, 7, 5, 10, 12, 5, 7}. The value 5 occurs more frequently than any other value (three times), making it the mode of the dataset.

- **Python Example:**

```
edian = np.median(data)
print("Median:", median)
```


d. Standard Deviation

- Standard deviation is a key measure in descriptive statistics that quantifies the amount of variation or dispersion in a dataset. It indicates how much individual data points deviate from the dataset's mean (average).
- Variance is a statistical measure that helps quantify the extent to which data points in a dataset differ from the mean, providing insights into the spread or dispersion of the data.
- In data science, variance is a concept that is closely related to the idea of variability or spread within a dataset. It is a key term in the context of machine learning, particularly when discussing model performance, bias-variance trade-off, and the generalization of models to new, unseen data.
- A **low standard deviation** means the data points are clustered closely around the mean, while a **high standard deviation** indicates that the data points are spread out over a wider range of values.

For a dataset with n values $(x_1, x_2, \dots, x_n)$:

1. Calculate the mean (μ):

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

2. Compute the variance (σ^2):

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

3. The standard deviation (σ) is the square root of the variance:

$$\sigma = \sqrt{\sigma^2}$$

Need of Standard Deviation

1. **Measure of Variability:** It helps to understand the spread of data, which is crucial for tasks like normalization and feature scaling in machine learning.
2. **Outlier Detection:** A high standard deviation may indicate the presence of outliers.
3. **Comparison of Datasets:** Standard deviation enables comparing the variability between two or more datasets.

4. **Foundation for Further Analysis:** Used in hypothesis testing, z-scores, and confidence intervals.

Example: Standard Deviation Calculation

Suppose we have the dataset representing the test scores of 5 students:

Scores: 50,60,70,80,90

Step 1: Calculate the Mean

$$\mu = \frac{50 + 60 + 70 + 80 + 90}{5} = 70$$

Step 2: Find the Squared Differences

$$(50 - 70)^2 = 400, (60 - 70)^2 = 100, (70 - 70)^2 = 0, (80 - 70)^2 = 100, (90 - 70)^2 = 400$$

Step 3: Calculate Variance

$$\text{Variance } \sigma^2 = \frac{400 + 100 + 0 + 100 + 400}{5} = 200$$

Step 4: Calculate Standard Deviation

$$\sigma = \sqrt{200} \approx 14.14$$

The standard deviation of approximately 14.1414.1414.14 means that, on average, the test scores deviate from the mean (70) by 14.14 points.

Python Code:

```
import numpy as np
scores = [50, 60, 70, 80, 90]
std_dev = np.std(scores)
print("Standard Deviation:", std_dev)
```

Basic Visualization Techniques:

- Data visualization provides an important suite of tools for identifying a qualitative understanding.
- This can be helpful when we try to explore the dataset and extract some information to know about a dataset and can help with identifying patterns, corrupt data, outliers, and much more.
- If we have a little domain knowledge, then data visualizations can be used to express and identify key relationships in plots and charts that

are more helpful to yourself and stakeholders than measures of association or significance.

Matplotlib

Matplotlib is a widely used library for creating static, animated, and interactive visualizations in Python. It provides a variety of charts, such as line charts, scatter charts, pie charts, and subplots. Here's an explanation of each of these components, along with examples.

1. Components of a Chart

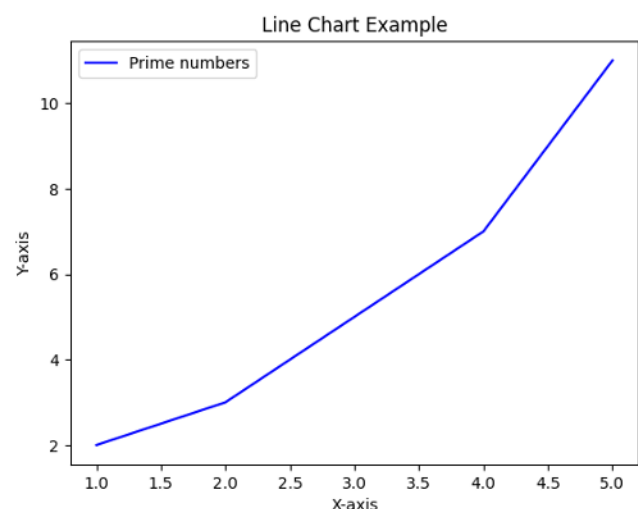
A typical chart in Matplotlib consists of several components:

- **Title:** A description of what the chart represents.
- **Axes:** The x and y axes that define the chart's coordinate system.
- **Labels:** The axis labels that describe the data, such as "Time" or "Price."
- **Grid:** Optional lines that help make the chart easier to read.
- **Legend:** A key to explain the colors or symbols used in the chart.

2. Line Chart

A **line chart** displays information as a series of data points connected by straight line segments. It is useful for showing trends over time.

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [2, 3, 5, 7, 11]
plt.plot(x, y, label="Prime numbers",
color='blue')
plt.title('Line Chart Example')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.legend()
plt.show()
```

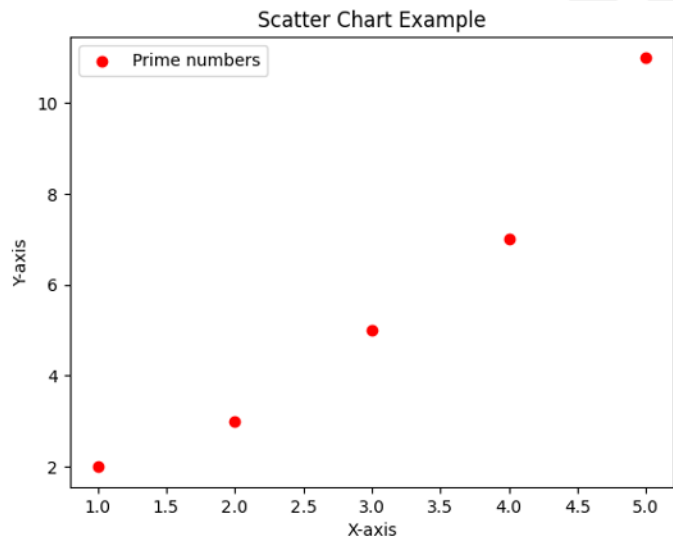


3. Scatter Chart

A **scatter plot** is a type of data visualization used to show the relationship between two continuous variables. It represents data points on a two-dimensional plane, where the x-axis corresponds to one variable and the y-axis corresponds to the other. Scatter plots are helpful for identifying

patterns, trends, correlations, or potential outliers in the data. A **scatter chart** is used to show the relationship between two variables.

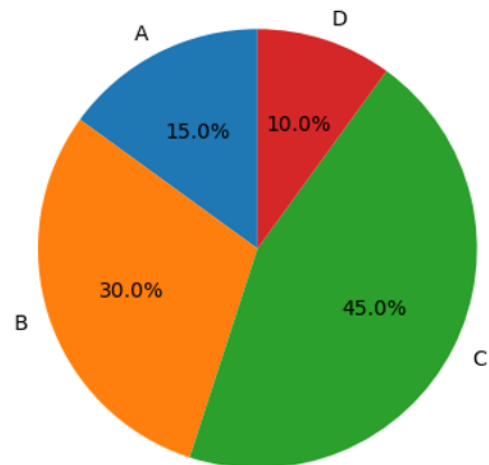
```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [2, 3, 5, 7, 11]
plt.scatter(x, y, color='red', label="Prime numbers")
plt.title('Scatter Chart Example')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.legend()
plt.show()
```



4. Pie Chart

A **pie chart** is used to represent data as slices of a circle, showing relative proportions of a whole.

```
import matplotlib.pyplot as plt
labels = ['A', 'B', 'C', 'D']
sizes = [15, 30, 45, 10]
plt.pie(sizes, labels=labels,
autopct='%1.1f%%',
startangle=90)plt.title('Pie Chart Example')
plt.show()
```



5. Subplots

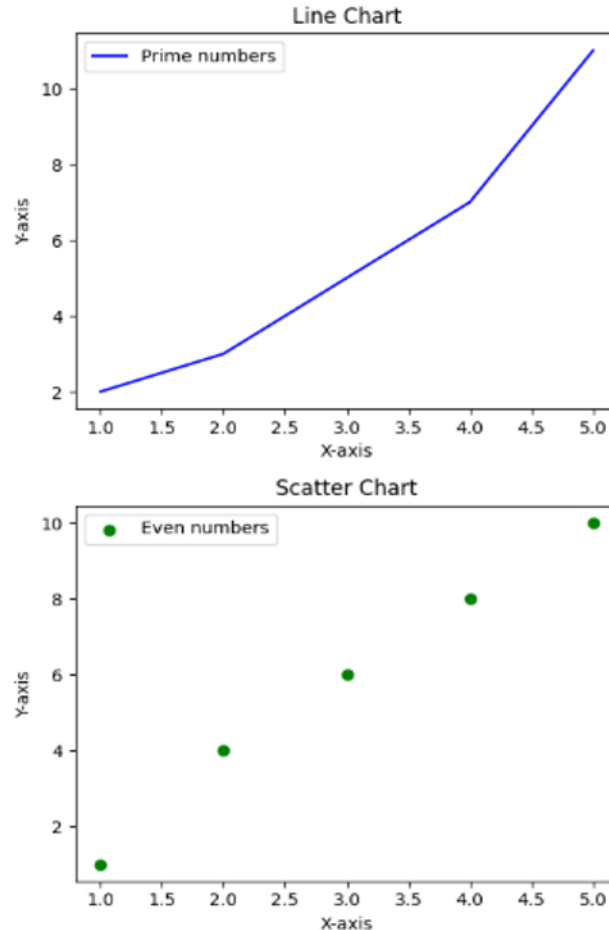
Subplots allow you to create multiple charts in a single figure, arranged in a grid. This is useful for comparing multiple datasets or visualizing different aspects of the same dataset.

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
```

```

y1 = [2, 3, 5, 7, 11]
y2 = [1, 4, 6, 8, 10]
fig, axs = plt.subplots(2, 1, figsize=(5, 8)) # 2 rows, 1 column
axs[0].plot(x, y1, label='Prime numbers', color='blue')
axs[0].set_title('Line Chart')
axs[0].set_xlabel('X-axis')
axs[0].set_ylabel('Y-axis')
axs[0].legend()
axs[1].scatter(x, y2, label='Even numbers', color='green')
axs[1].set_title('Scatter Chart')
axs[1].set_xlabel('X-axis')
axs[1].set_ylabel('Y-axis')
axs[1].legend()
plt.tight_layout() # Adjust the layout to prevent overlap
plt.show()

```



6. Histogram

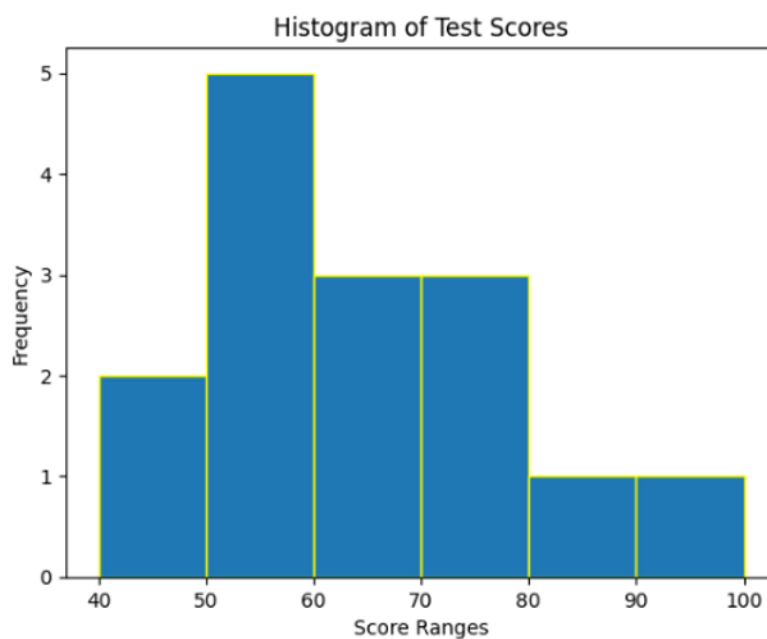
Histogram is a type of graphical representation used in statistics to show the distribution of numerical data. It looks somewhat like a bar chart, but unlike bar graphs, which are used for categorical data, histograms are designed for continuous data, grouping it into logical ranges which are also known as "bins."

It is used in data visualization to summarize and display the frequency of data points within specified ranges (called bins or intervals). Histograms are particularly useful for understanding the shape, central tendency, variability, and spread of a dataset.

Code:

```
import matplotlib.pyplot as plt
test_scores = [45, 50, 52, 55, 56, 60, 62, 45, 77, 51, 72, 62, 78, 90, 82]
plt.hist(test_scores, bins=[40, 50, 60, 70, 80, 90, 100], edgecolor='yellow')
plt.xlabel('Score Ranges')
plt.ylabel('Frequency')
plt.title('Histogram of Test Scores')
plt.show()
```

Output:



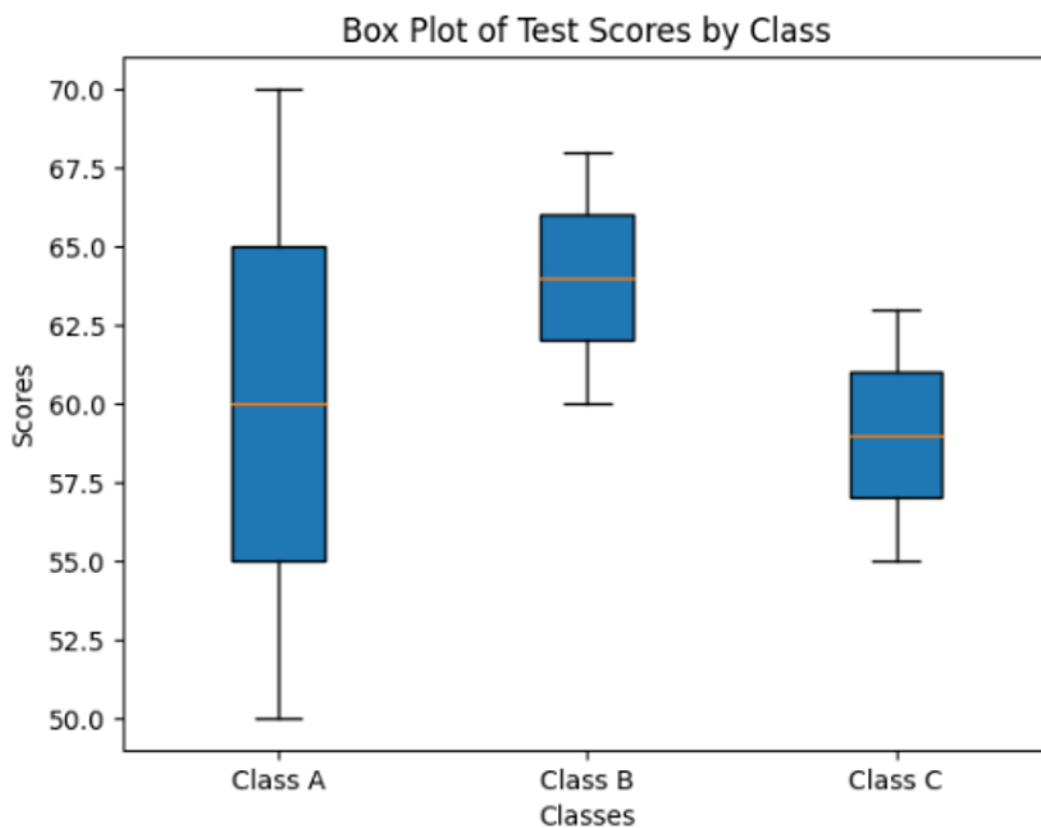
7. Box Plot

A **box plot** (or box-and-whisker plot) is a graphical representation used to summarize the distribution of a dataset. It highlights key statistical measures such as the median, quartiles, and potential outliers. Box plots are particularly useful for comparing distributions between different groups.

```
import matplotlib.pyplot as plt

class_a_scores = [50, 55, 60, 65, 70]
class_b_scores = [60, 62, 64, 66, 68]
class_c_scores = [55, 57, 59, 61, 63]

# Combine data into a single list
data = [class_a_scores, class_b_scores, class_c_scores]
plt.boxplot(data, labels=['Class A', 'Class B', 'Class C'], patch_artist=True)
plt.xlabel('Classes')
plt.ylabel('Scores')
plt.title('Box Plot of Test Scores by Class')
plt.show()
```

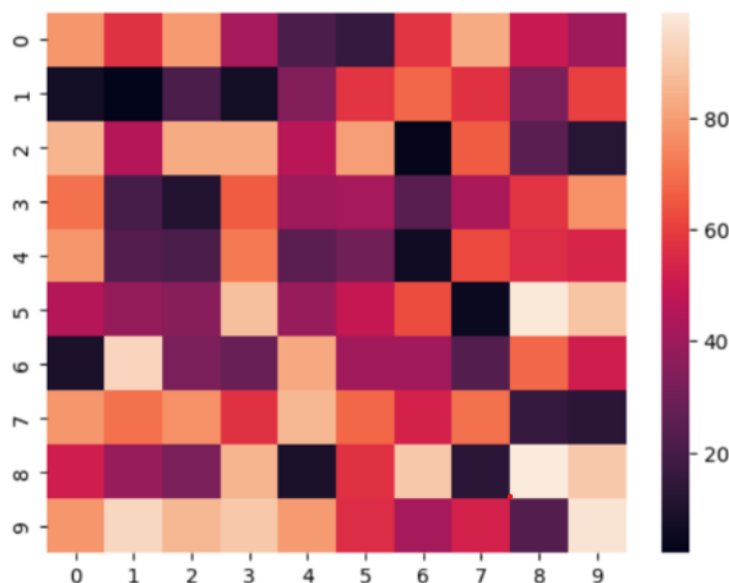


8. Heat map

- Heatmap is defined as a graphical representation of data using colors to visualize the value of the matrix. In this, to represent more common values or higher activities brighter colors basically reddish colors are used and to represent less common or activity values, darker colors are preferred.
- Heatmap is also defined by the name of the shading matrix. Heatmaps in Seaborn can be plotted by using the `seaborn.heatmap()` function.
- A heatmap (aka heat map) depicts values for a main variable of interest across two axis variables as a grid of colored squares. The axis variables are divided into ranges like a bar chart or histogram, and each cell's color indicates the value of the main variable in the corresponding cell range.

Code:

```
import numpy as np
import seaborn as sn
import matplotlib.pyplot as plt
# generating 2-D 10x10 matrix of random numbers from 1 to 100
data = np.random.randint(low = 1, high = 100, size = (10, 10))
print("The data to be plotted:\n")
print(data)
hm = sn.heatmap(data = data)
plt.show()
```

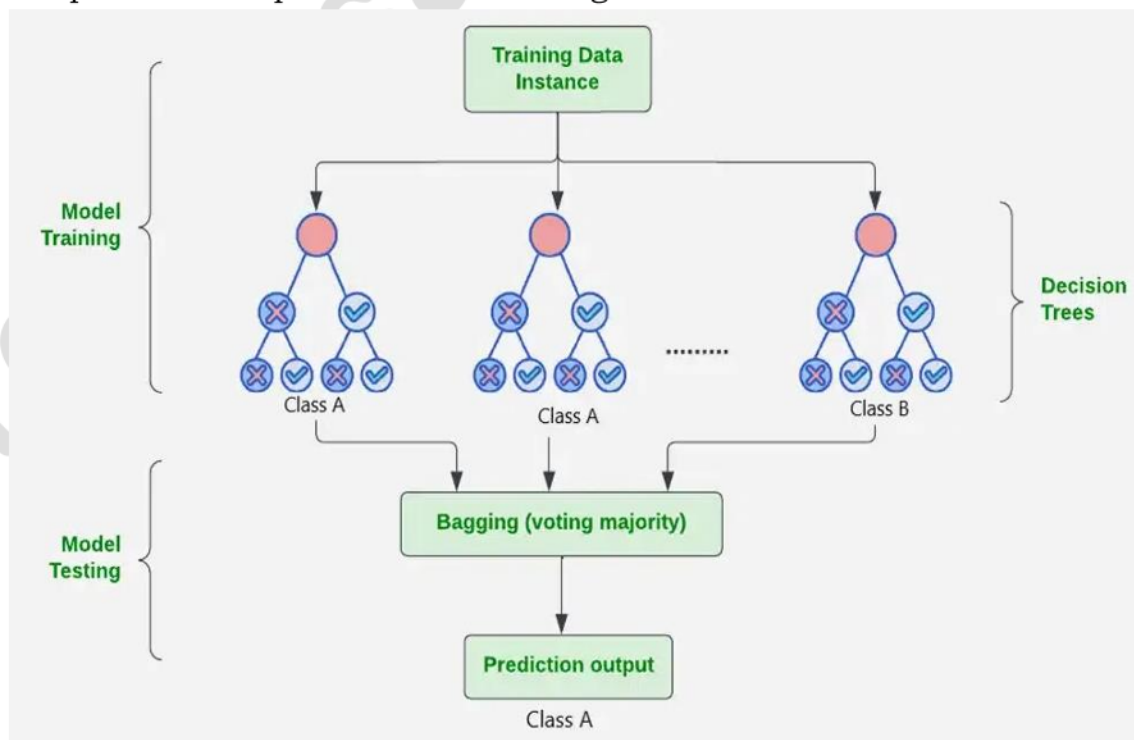


Chapter No: 07 Introduction to Machine Learning**Chapter No: 08 Regression Analysis****Chapter No: 10 Machine Learning Algorithms**

Note: for above chapters refer TYCS AI Notes.

Random forest Algorithm

- Random Forest is a popular machine learning algorithm that belongs to the supervised learning technique.
- It can be used for both Classification and Regression problems in ML.
- It is based on the concept of ensemble learning, which is a process of combining multiple classifiers to solve a complex problem and to improve the performance of the model.
- Random Forest is a classifier that contains a number of decision trees on various subsets of the given dataset and takes the average to improve the predictive accuracy of that dataset.
- Instead of relying on one decision tree, the random forest takes the prediction from each tree and based on the majority votes of predictions, and it predicts the final output.
- It takes different random parts of the dataset to train each tree and then it combines the results by averaging them. This approach helps improve the accuracy of predictions.
- The greater number of trees in the forest leads to higher accuracy and prevents the problem of overfitting.



Key Features of Random Forest

- Automatically handles missing values during training, eliminating the need for manual imputation.
- Algorithm ranks features based on their importance in making predictions offering valuable insights for feature selection and interpretability.
- Scales Well with Large and Complex Data without significant performance degradation.
- Algorithm is versatile and can be applied to both classification tasks (e.g., predicting categories) and regression tasks (e.g., predicting continuous values).

How Random Forest Algorithm Works?

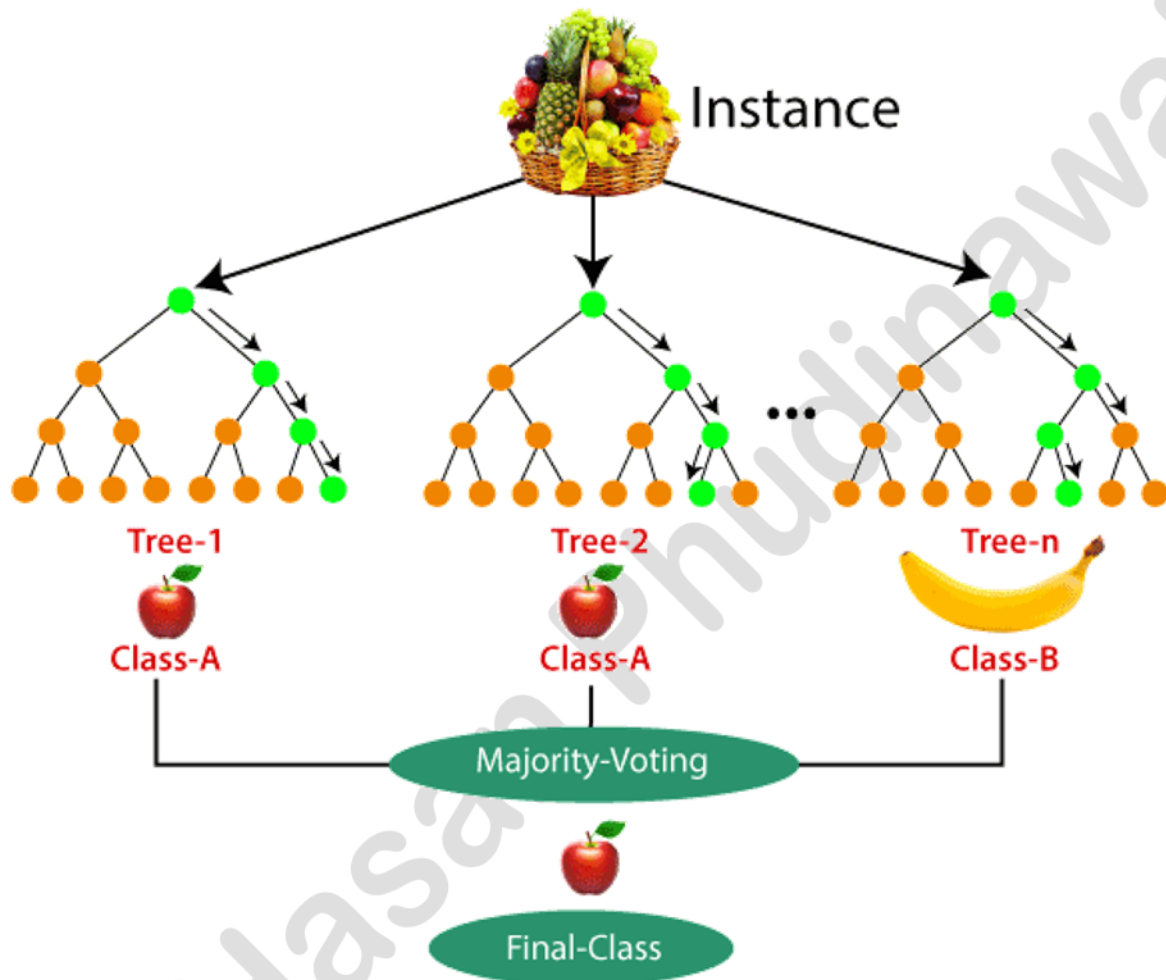
- Random Forest builds multiple decision trees using random samples & features to reduce mistakes of the data. Each tree is trained on a different subset of the data which makes each tree unique i.e. Every tree in the forest makes its own predictions without relying on others.
- When creating each tree the algorithm randomly selects a subset of features or variables to split the data rather than using all available features at a time. This adds diversity to the trees.
- Each decision tree in the forest makes a prediction based on the data it was trained on. When making final prediction random forest combines the results from all the trees.
- For classification tasks the final prediction is decided by a majority vote. This means that the category predicted by most trees is the final prediction.
- For regression tasks the final prediction is the average of the predictions from all the trees.
- The randomness in data samples and feature selection helps to prevent the model from overfitting making the predictions more accurate and reliable.
- Different predictions improve accuracy

Advantages of Random Forest

- Random Forest provides very accurate predictions even with large datasets.
- Random Forest can handle missing data well without compromising with accuracy. Combining the predictions from different trees leads to a more accurate final result.
- It doesn't require normalization or standardization on dataset.
- When we combine multiple decision trees it reduces the risk of overfitting of the model.

Limitations of Random Forest

- It can be computationally expensive especially with a large number of trees.
- It's harder to interpret the model compared to simpler models like decision trees.



Ensemble Learning

- Ensemble learning is primarily used to improve the model performance, such as classification, prediction, function approximation, etc.
- The ensemble methods in machine learning combine the insights obtained from multiple learning models to facilitate accurate and improved decisions. In learning models, noise, variance, and bias are the major sources of error. The ensemble methods in machine learning help minimize these error-causing factors, thereby ensuring the accuracy and stability of machine learning (ML) algorithms.
- Example : Assume that you are developing an app for the travel industry.

- It is obvious that before making the app public, you will want to get crucial feedback on bugs and potential loopholes that are affecting the user experience.
- What are your available options for obtaining critical feedback?
 - 1) Soliciting opinions from your parents, spouse, or close friends.
 - 2) Asking your co-workers who travel regularly and then evaluating their response.
 - 3) Rolling out your travel and tourism app in beta to gather feedback from non-biased audiences and the travel community.
- Think for a moment about what you are doing.
- You are taking into account different views and ideas from a wide range of people to fix issues that are limiting the user experience.
- The ensemble neural network and ensemble algorithm do precisely the same thing.
- Example: Imagine a group of blindfolded people playing the touch-and-tell game, where they are asked to touch and explore a mini donut factory that no one of them has ever seen before.
- Since they are blindfolded, their version of what a mini donut factory looks like will vary, depending on the parts of the appliance they touch.
- Now, suppose they are personally asked to describe what they touched.
- In that case, their individual experiences will give a precise description of specific parts of the mini donut factory.
- Still, collectively, their combined experiences will provide a highly detailed account of the entire equipment.
- Similarly, ensemble methods in machine learning employ a set of models and take advantage of the blended output, which, compared to a solitary model, will most certainly be a superior option when it comes to prediction accuracy.
- There are 3 most common ensemble learning methods in machine learning.
 - Bagging
 - Boosting
 - Stacking

Bagging

- Bagging is a method of ensemble modelling, which is primarily used to solve supervised machine learning problems.
- It is generally completed in two steps as follows:
- **Bootstrapping:** It is a random sampling method that is used to derive samples from the data using the replacement procedure. In this method, first, random data samples are fed to the primary model, and then a base learning algorithm is run on the samples to complete the learning process.

- **Aggregation:** This is a step that involves the process of combining the output of all base models and, based on their output, predicting an aggregate result with greater accuracy and reduced variance.
- **Example:** In the Random Forest method, predictions from multiple decision trees are ensembled parallelly. Further, in regression problems, we use an average of these predictions to get the final output, whereas, in classification problems, the model is selected as the predicted class.

Boosting

- Boosting is an ensemble method that enables each member to learn from the preceding member's mistakes and make better predictions for the future. Unlike the bagging method, in boosting, all base learners (weak) are arranged in a sequential format so that they can learn from the mistakes of their preceding learner.
- Hence, in this way, all weak learners get turned into strong learners and make a better predictive model with significantly improved performance.
- We have a basic understanding of ensemble techniques in machine learning and their two common methods, i.e., bagging and boosting.

Chapter No: 09

Model Evaluation and Selection

Cross-validation: k-fold cross-validation, stratified cross-validation, Hyperparameter tuning and model selection

Accuracy:

Formula: $(TP + TN) / (TP + TN + FP + FN)$

- Accuracy measures the overall correctness of the classifier and is the ratio of correctly predicted instances to the total instances.

Precision (Positive Predictive Value):

Formula: $TP / (TP + FP)$

- Precision measures the accuracy of the positive predictions and is the ratio of correctly predicted positive observations to the total predicted positives.

Recall (Sensitivity, True Positive Rate):

- Formula: $\text{Recall} = \text{True Positives} / (\text{True Positive} + \text{False Negative})$
- Recall measures the ability of the classifier to capture all the positive instances and is the ratio of correctly predicted positive observations to the total actual positives.
- Recall, also known as sensitivity or true positive rate, is a key metric in the field of data mining, particularly in classification tasks.
- It is especially relevant when the goal is to identify as many positive instances as possible, even if it means accepting a higher number of false positives.
- The ability of a model to find all the relevant cases within a data set. Mathematically, we define recall as the number of true positives divided by the number of true positives plus the number of false negatives.
- Precision: The ability of a classification model to identify only the relevant data points.
- Recall is defined as the ratio of true positives to the sum of true positives and false negatives:

F1 Score:

Formula: $2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$

- The F-measure, also known as the F1 score, is a metric commonly used in data mining and machine learning to assess the performance of a classification model.
- It combines precision and recall into a single metric, providing a balanced measure of a model's effectiveness.
- The F-measure is especially useful when there is an uneven class distribution, and you want to consider both false positives and false negatives.
- The F1 score is the harmonic mean of precision and recall. It provides a balance between precision and recall and is especially useful when there is an uneven class distribution.

Specificity (True Negative Rate):

Formula: $TN / (TN + FP)$

- Specificity measures the ability of the classifier to correctly identify negative instances.

Confusion Matrix:

- A confusion matrix is a fundamental tool in model validation, particularly for evaluating the performance of classification models.
- It provides a detailed breakdown of the predictions made by a model, allowing you to assess how well the model is performing in terms of true positives, true negatives, false positives, and false negatives.
- These metrics are essential for calculating various performance metrics such as accuracy, precision, recall, F1 score, and others.
- Here is a breakdown of the elements in a confusion matrix:

True Positives (TP):

- Instances that are actually positive and are correctly predicted as positive by the model.

True Negatives (TN):

- Instances that are actually negative and are correctly predicted as negative by the model.

False Positives (FP):

- Instances that are actually negative but are incorrectly predicted as positive by the model.

False Negatives (FN):

- Instances that are actually positive but are incorrectly predicted as negative by the model.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

		true class			
		EFR	LFR	total	
predicted class	EFR	True Positives (TP)	False Positives (FP)	predicted EFR	$PR = \frac{TP}{TP+FP}$
	LFR	False Negatives (FN)	True Negatives (TN)	predicted LFR	$RE = \frac{TP}{TP+FN}$
		true EFR	true LFR		$CA = \frac{TP+TN}{TP+TN+FP+FN}$
					$F_1 = \frac{2TP}{2TP+FP+FN}$

Cross-Validation

- Cross-validation is a technique used in model validation to assess the performance of a machine learning model.
- It helps to ensure that the model generalizes well to new, unseen data by evaluating its performance across multiple subsets of the dataset.
- The primary goal is to provide a more robust estimate of the model's performance compared to a single train-test split.
- Here are the key concepts and steps involved in cross-validation:
- In traditional model validation, the dataset is split into two subsets: a training set used for model training and a testing set used for evaluation. However, a single split may lead to variability in performance evaluation, depending on which instances end up in the test set.

K-Fold Cross-Validation

- In k-fold cross-validation, the dataset is divided into k equally-sized folds. The model is trained and evaluated k times, each time using a different fold as the test set and the remaining folds as the training set. This process results in k performance metrics, which are then averaged to obtain a more reliable estimate of the model's performance.

Steps in K-Fold Cross-Validation:**1. Dataset Splitting:**

- Divide the dataset into k folds. Common choices for k are 5 or 10, but it can vary based on the dataset size and computational resources.

2. Training and Evaluation:

- Perform k iterations. In each iteration, use k-1 folds for training and the remaining fold for testing. Train the model on the training set and evaluate its performance on the test set.

3. Performance Metrics:

- Calculate performance metrics (e.g., accuracy, precision, recall, F1 score) for each iteration.

4. Average Metrics:

- Average the performance metrics across all k iterations to obtain a more stable estimate of the model's performance.

Stratified k-fold cross-validation

- This technique is similar to k-fold cross-validation with some little changes. This approach works on stratification concept, it is a process of rearranging the data to ensure that each fold or group is a good representative of the complete dataset. To deal with the bias and variance, it is one of the best approaches.
- It can be understood with an example of housing prices, such that the price of some houses can be much high than other houses. To tackle such situations, a stratified k-fold cross-validation technique is useful.

Hyperparameter Tuning

- Hyperparameter tuning refers to the process of finding the best combination of hyperparameters for a machine learning model to optimize its performance on a given dataset.
- Hyperparameters are settings that are not learned during training but are set before the learning process begins.
- Examples include the learning rate, number of hidden layers, number of neurons per layer, dropout rate, regularization strength, and the type of kernel in Support Vector Machines.

Key Aspects of Hyperparameter Tuning:

Improve model performance by finding the most effective hyperparameter configuration.

1. **Techniques:**

- **Grid Search:** Systematically searches through a predefined set of hyperparameter values.
- **Random Search:** Samples random combinations of hyperparameters within a specified range.
- **Bayesian Optimization:** Uses probabilistic models to predict the best hyperparameter combination.
- **Hyperband:** Combines random search and early stopping to efficiently evaluate hyperparameter configurations.

2. **Validation Strategy:** Typically involves using cross-validation to ensure the selected hyperparameters generalize well on unseen data.

Model Selection

Model selection involves choosing the most suitable algorithm or model architecture for a specific task or dataset. The goal is to identify the model that achieves the best balance between complexity and performance.

Key Aspects of Model Selection:

- **Performance Metrics:** Metrics such as accuracy, precision, recall, F1 score, mean squared error (MSE), or AUC-ROC are used to compare models.
- **Bias-Variance Tradeoff:** The selected model should generalize well, avoiding underfitting and overfitting.

2. **Considerations:**

- Data characteristics, such as size, dimensionality, and distribution.
- Computational resources and time constraints.
- Domain-specific requirements, such as interpretability or deployment constraints.

Relationship Between Hyperparameter Tuning and Model Selection

Hyperparameter tuning and model selection are closely related but distinct processes.

- **Model selection** is the broader task of identifying the best algorithm or architecture.
- **Hyperparameter tuning** fine-tunes the chosen model to maximize its performance.